

Hotel Booking Demand: Classification and Prediction of Cancellations

Team 19: Daniela Lorriz Bautista¹, Kevin Davide Dextre Gonzales¹, Matteo Fumagalli¹

¹*MSc in Data Science, University of Milano-Bicocca*

January 2026

Abstract

This study investigates the Hotel Booking Demand dataset, publicly available on Kaggle, with the ultimate goal of identifying the most effective classification models for predicting reservation cancellations. The analysis is implemented through a complete machine learning pipeline developed in KNIME and follows a structured workflow consisting of data preparation, feature engineering and selection, model training and evaluation. Several classifiers are trained and compared using multiple evaluation metrics to assess their ability to correctly identify reservations with a higher probability of cancellation. Overall, the results indicate that, in this context, boosting-based models achieve strong predictive performance, characterized by high accuracy and well-calibrated probability estimates.

Keywords: hotel booking, cancellation prediction, machine learning, classification, boosting techniques

Contents

1	Introduction	1
2	Dataset	2
2.1	Structure	2
2.2	Data Exploration	2
3	Preprocessing	3
3.1	Feature Engineering	3
3.2	Feature Selection	3
4	Classification	4
4.1	Methodology	4
4.2	Models	4
4.3	Metrics	5
5	Results	5
6	Conclusion	6

1 Introduction

Hotels handle a large volume of reservations on a daily basis, including bookings related to business travels, leisure activities, and vacations. A considerable portion of these reservations is eventually cancelled, often resulting in revenue losses and inefficient use of

available resources, as cancelled rooms cannot always be reassigned in time. According to industry reports [1], hotel booking cancellation rates typically range between 18% and 42%, highlighting the relevance of this issue within revenue management strategies. Traditional approaches to managing cancellations are frequently based on manual assessments and managerial experience. While valuable, such methods may lack predictive accuracy and can negatively affect customer satisfaction. This project aims to address this challenge by analyzing the Hotel Booking Demand dataset and developing a set of classification models capable of estimating the probability that a reservation will be cancelled. Accurately identifying bookings with a higher risk of cancellation allows hotels to implement targeted countermeasures, such as prepayment policies, booking guarantees, or reminder communications, thereby reducing potential financial losses and improving operational efficiency.

The remainder of this report is organized as follows. Section 2 describes the dataset and presents an exploratory analysis of its main characteristics. Section 3 outlines the preprocessing steps, including feature engineering and feature selection. Then, Section 4 introduces the classification task, detailing the methodology, the models, and the evaluation metrics. Section 5 reports and discusses the main results. Finally, Section 6 concludes the report and highlights possible future work.

2 Dataset

The dataset used [2] contains booking information for two different hotels: a resort hotel and a city hotel. It consists of a total of 119,390 records, each of which corresponds to a single hotel reservation. The data covers a time period ranging from July 2015 to August 2017. Therefore, the dataset has not been updated beyond this interval.

2.1 Structure

The dataset comprises 32 attributes describing various aspects of hotel bookings. The target variable is `is_canceled`, which indicates whether a reservation was cancelled or not. In particular, the attributes included in the dataset are listed below, along with a brief description of each variable:

- **hotel:** H1 = Resort Hotel, H2 = City Hotel.
- **is_canceled:** 0 = not cancelled, 1 = cancelled.
- **lead_time:** Number of days between booking date and arrival date.
- **arrival_date_year:** Year of arrival.
- **arrival_date_month:** Month of arrival.
- **arrival_date_week_number:** Week number of arrival.
- **arrival_date_day_of_month:** Day of arrival.
- **stays_in_weekend_nights:** Number of weekend nights booked.
- **stays_in_week_nights:** Number of weekday nights booked.
- **adults:** Number of adults.
- **children:** Number of children.
- **babies:** Number of babies.
- **meal:** BB = Bed and Breakfast, HB = Half Board, FB = Full Board, Undefined/SC = no meal.
- **country:** Country of origin encoded using ISO 3166-1 alpha-3 codes (e.g., PRT = Portugal, GBR = United Kingdom).
- **market_segment:** TA = Travel Agents, TO = Tour Operators, Corporate = Business customers, Direct = Direct bookings, Groups = Group bookings, Offline TA/TO = Offline travel agents or tour operators, Undefined = Not defined.
- **distribution_channel:** TA = Travel Agents, TO = Tour Operators, Direct = Direct bookings, Corporate = Corporate / business bookings, GDS = Global Distribution System.
- **is_repeated_guest:** 0 = not repeated, 1 = repeated.
- **previous_cancellations:** Number of previous cancelled bookings.
- **previous_bookings_not_canceled:** Number of previous non-cancelled bookings.
- **reserved_room_type:** Room type reserved at booking time.
- **assigned_room_type:** Room type assigned at check-in.
- **booking_changes:** Number of changes made to the booking.
- **deposit_type:** No Deposit = no deposit made, Non Refund = non-refundable deposit, Refundable = refundable deposit.
- **agent:** Booking agent identifier.
- **company:** Company identifier.
- **days_in_waiting_list:** Number of days on the waiting list.
- **customer_type:** Contract = booking with an allotment or other contract arrangement, Group = group booking, Transient = individual booking not part of a group or contract, Transient-Party = individual booking as part of a group or party.
- **adr:** Average Daily Rate.
- **required_car_parking_spaces:** Number of parking spaces requested.
- **total_of_special_requests:** Number of special requests.
- **reservation_status:** Canceled = booking was cancelled by the customer, Check-Out = customer has checked out after their stay, No-Show = customer did not arrive for the booking.
- **reservation_status_date:** Date of the last reservation status update.

2.2 Data Exploration

The dataset includes a mix of categorical and numerical variables. Categorical attributes mainly describe qualitative aspects of the booking, the customer, and the sales channel, while numerical variables provide quantitative information related to timing and stay duration. Additionally, the target variable `is_canceled` is binary, making the task a binary classification problem.

An initial exploratory analysis revealed the presence of missing values in the `children`, `agent`, and `company` attributes. Missing values in the `children` variable, observed in only four records, were replaced with the

value 0, under the assumption that the absence of information indicates that no children were included in the reservation. Instead, the **agent** and **company** attributes contained a large portion of missing values and represented identifier codes with limited predictive relevance; therefore, both features were removed from the dataset. Similarly, attributes related to reservation outcomes, namely **reservation_status** and **reservation_status_date**, were excluded, as they provide information that becomes available only after the booking decision. As shown in Figure 1, the dataset is moderately imbalanced, with a higher proportion of non-cancelled bookings compared to cancelled ones. However, the imbalance was not considered severe enough to require explicit resampling or balancing techniques, and the dataset was found to be suitable for training classification models.

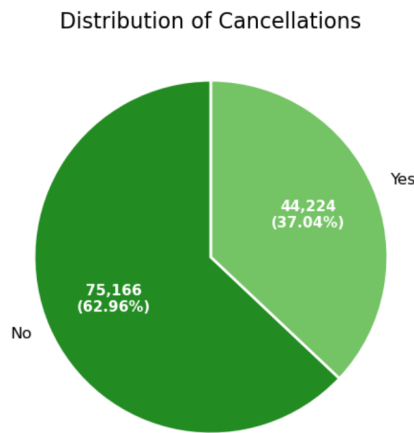


Figure 1: Distribution of cancellations

During the exploratory analysis, some numerical variables showed a limited presence of extreme values. Instead of performing outlier removal techniques, potential effects were implicitly handled during the pre-processing phase. A further inspection of the data distribution after feature engineering did not reveal problematic behaviours likely to affect the analysis.

3 Preprocessing

Through the preprocessing phase, the original data was transformed into a structured format suitable for machine learning algorithms, with the aim of enhancing their predictive power while reducing redundancy.

3.1 Feature Engineering

Some domain-specific transformations were applied to derive more meaningful and informative features to predict the target variable **is_canceled**:

- **Geographical aggregation:** the **country** field was mapped to its respective **continent** using the **country_converter** library to reduce high-cardinality noise.

- **Temporal features:** the binary variable **is_weekend** was derived to indicate whether the arrival date falls on a weekend, as well as **season** to capture seasonal trends in hotel bookings. Specifically, **arrival_date_month**, **arrival_date_week_number**, and **arrival_date_day_of_month** were used for derivation and subsequently dropped to reduce dimensionality, while only **arrival_date_year** was maintained to capture long-term trends.
- **Guest composition:** the overall feature **total_guests** aggregated **adults**, **babies**, and **children**, simplifying the representation while retaining essential information.
- **Stay duration:** the overall feature **total_nights** combined **stays_in_week_nights** and **stays_in_weekend_nights**, obtaining the total length of stay.
- **Room allocation:** the binary indicator **room_match** was defined comparing **assigned_room_type** with **reserved_room_type**, capturing whether guests received their requested room type, which may influence cancellation decisions.

After creating these derived features, the original ones were removed to prevent multicollinearity and reduce model complexity. Furthermore, before searching for the optimal subset of features, the correlation matrix for all numerical variables was computed to detect potential dependencies. The analysis did not reveal strong linear correlations among these features, validating the feature engineering approach. Instead, for categorical variables, the co-occurrence patterns were examined to identify redundant information. In particular, the analysis of the relationship between **market_segment** and **distribution_channel** revealed substantial overlap in the information they capture. To prevent redundancy and reduce dimensionality, **distribution_channel** was removed from the feature set, retaining **market_segment** as it directly explains customer booking behaviour.

3.2 Feature Selection

A systematic feature selection process, based on feature importance derived from a **Random Forest** classifier, was implemented to identify the most relevant set of features for predicting cancellations. The aim was to improve model efficiency and interpretability without reducing classification performance.

Prior to modelling, the target variable **is_canceled** was mapped from $\{0, 1\}$ to $\{\text{"No"}, \text{"Yes"}\}$ for improved interpretability during analysis and visualization, whereas all multi-class categorical features were encoded using the one-hot encoding strategy, creating a binary indicator for each category. Then, a stratified 80-20 train-test split was performed, preserving the

distribution of the outcome variable `is_canceled` in both subsets. To avoid data leakage, feature selection was based exclusively on the training set, employing a **Random Forest** classifier and calculating feature importance using a custom score related to the depth at which explanatory variables were used for splitting in the decision trees.

The importance metric was defined as:

$$\begin{aligned} \text{importance} = & (\text{l}_0_splits \times 3) \\ & + (\text{l}_1_splits \times 2) \\ & + (\text{l}_2_splits \times 1) \end{aligned}$$

where `lx_splits` represents the number of times a feature was used for splitting at depth `X` across all trees in the forest. This weighted schema reflects the hierarchical importance of tree splits: features selected at earlier depths capture more relevant decision boundaries, thus having a greater overall impact on predictions. By assigning decreasing weights to splits at increasing depths, prioritized features were those identified as the most discriminative early in the decision process. Features were ranked in descending order of importance, and both cumulative importance and relative cumulative importance were computed. As illustrated in Figure 2, visual analysis of the cumulative importance curve revealed a notable flattening beyond the 90% threshold, indicating a decreasing marginal contribution from the introduction of additional variables. The set of features to exclude was obtained by considering whether all categories of a specific variable fell entirely within the bottom 10% region of cumulative importance. Following this methodology, `hotel`, `meal`, `country`, and `season` were removed from the optimal subset of features. Indeed, these variables collectively accounted for less than 10% of the total importance score while adding unnecessary complexity to the model.

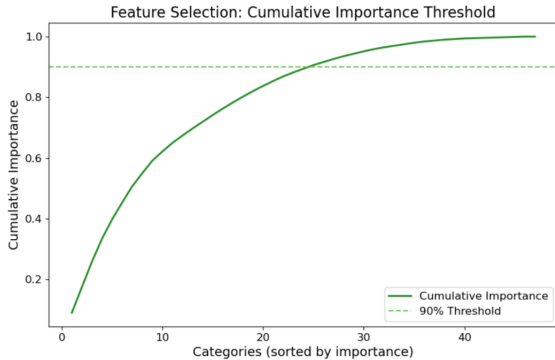


Figure 2: Cumulative Feature Importance Curve

4 Classification

The booking cancellation prediction task was addressed using a supervised binary classification framework, in order to predict whether a hotel booking

would be cancelled based on the engineered and selected features. This section outlines the adopted methodology, the classification models evaluated, and the metrics used for performance assessment.

4.1 Methodology

All classification models were trained and evaluated using a unified workflow to ensure methodological consistency across classifiers and to enable a fair comparison of their performance. Following preprocessing and feature selection, the modelling workflow was structured into two main phases: hyperparameter tuning, and model training and evaluation.

Hyperparameter Tuning Phase. In the first stage, hyperparameter optimization was performed for tree-based classifiers using a stratified 80-20 training-validation split derived from the initial training data. A parameter optimization loop explored candidate configurations and selected the optimal setting based on validation performance. Instead, **Logistic Regression** did not require hyperparameter tuning and was trained directly after applying feature normalization to account for differences in feature scale.

Model Training and Evaluation Phase. In the second stage, the optimal configurations were evaluated using 5-fold cross-validation on the training set to obtain more reliable performance estimates across different data splits [3]. Performance metrics were aggregated across folds to obtain robust performance estimates. After cross-validation, each classifier was retrained on the full training set and subsequently evaluated on the holdout test set. This procedure was adopted to determine performance estimates that better reflect generalization to unseen data while maintaining consistency across models.

Throughout the process, models produced both class predictions and associated probability estimates for the positive class, which were aggregated and later used for metrics computation and discussion.

4.2 Models

Multiple classification algorithms were considered to compare approaches with different levels of complexity and learning capacity, including a linear baseline and several tree-based and ensemble methods. The aim was to identify the model that best captures patterns related to booking cancellations while providing robust and generalizable predictions.

Logistic Regression was used as a baseline linear model. Its inclusion provides a simple reference, allowing for performance comparison against more complex non-linear approaches and assessing whether a linear decision boundary is sufficient to capture cancellation patterns.

Decision Tree model was employed to capture non-linear relationships between features using hierarchical decision rules. Hyperparameters explored included **pruning confidence** [0.1, 0.3] and **minimum number of instances per leaf** [20, 200]. Decision trees provide increased flexibility compared to linear classifiers, are relatively interpretable, and can reveal meaningful patterns in the data; although, they can be prone to instability.

Random Forest was included as an ensemble extension of decision trees, combining multiple models trained on bootstrap samples with random feature subsets at each split to improve generalization. This ensemble strategy is commonly adopted to mitigate the overfitting effects observed in individual decision trees [4]. The hyperparameter settings investigated were the **number of trees** [200, 400], **maximum tree depth** [10, 30], **column fraction per tree** [0.1, 1.0], and **minimum node size** [3, 5].

Gradient Boosted Trees was selected for its ability to iteratively correct errors of previous models, capturing complex interactions among features. Hyperparameter values examined for tuning included **number of boosting iterations** [100, 400], **learning rate** [0.01, 0.1], **maximum tree depth** [3, 6], and **minimum child size** [20, 80].

XGBoost was evaluated as a high-performance boosting algorithm capable of modelling complex decision boundaries and producing probability estimates that are often well suited for decision-oriented evaluation. Considered hyperparameters included **learning rate** [0.01, 0.1], **maximum depth** [3, 10], and **minimum child weight** [10, 50].

All tree-based and ensemble models were optimized using a random search technique to explore candidate configurations efficiently [5]. For **Gradient Boosted Trees** and **XGBoost**, the random search procedure was allowed up to 1,000 iterations due to their larger and more complex hyperparameter spaces; whereas for **Decision Tree** and **Random Forest**, the search was limited to 100 iterations. Each model was trained using the same feature set and evaluation workflow described previously, ensuring that differences in predictive performance reflect the characteristics of the learning algorithms rather than variations in preprocessing or feature selection.

4.3 Metrics

The performance of all classifiers was evaluated using a set of complementary metrics to provide a comprehensive assessment of their predictive capabilities on the test set. Both overall and positive-class-specific metrics were considered to capture how well models identify cancellations while also reflecting general classification performance. Overall metrics include

accuracy and Cohen’s kappa, while positive-class metrics comprise recall, precision, and F1-score, with specificity reported separately to capture performance on the negative class. The trade-off between recall and precision is particularly relevant in this context: recall quantifies the ability to detect actual cancellations, whereas precision assesses the reliability of predicted cancellations. The F1-Score summarizes these two measures into a single metric that captures the overall predictive performance for the positive class. Moreover, predicted probabilities for the positive class were also recorded to generate ROC curves, providing a visual assessment of the models’ discriminative ability.

In addition, log loss was computed to evaluate the quality of predicted probabilities, penalizing confident but incorrect predictions. For each instance, the log loss was calculated as:

$$\text{LogLoss}_i = -\left(y_i \cdot \ln(\hat{y}_i) + (1 - y_i) \cdot \ln(1 - \hat{y}_i)\right)$$

where $y_i \in \{0, 1\}$ denotes the true class label and $\hat{y}_i$ represents the predicted probability of the positive class. Predicted probabilities were constrained to lie between 10^{-15} and $1 - 10^{-15}$ to prevent numerical issues when computing the logarithm. Then, the overall log loss for a model was obtained by averaging the individual instance losses across the dataset.

The resulting metrics tables allowed for a direct comparison of models, while ROC curves and log loss values provided additional insights into probability calibration and discriminative ability, facilitating the identification of the optimal classifier for predicting booking cancellations.

5 Results

The classifiers’ performance was evaluated on the test set, comparing the metrics explained previously. In particular, Table 1 reports accuracy, F1-Score, and log loss for each model:

Model	Accuracy	F1-Score	Log Loss
Logistic Regression	0.813	0.712	0.419
Decision Tree	0.837	0.761	0.376
Random Forest	0.823	0.72	0.501
Gradient Boosted Trees	0.844	0.768	0.342
XGBoost	0.848	0.778	0.334

Table 1: Classification Metrics

All models achieved satisfactory performance, with accuracy scores ranging from 0.813 to 0.848, indicating that the feature engineering and selection pipeline effectively captured the relevant patterns in booking cancellations. In this scenario, it is essential to

strike a balance between high recall, which minimizes missed cancellations and supports effective revenue management, and high precision, which reduces false positives and helps preserve customer relations.

Among the classifiers, the simpler **Logistic Regression** was adopted as a baseline model, and the superior performance of tree-based and boosting methods confirms the presence of more complex underlying decision boundaries. Overall, **XGBoost** emerged as the top-performing model, achieving the highest accuracy and F1-Score, along with the lowest log loss. Therefore, this model not only produced accurate predictions but also provided well-calibrated probability estimates and robust discriminative capability. **Gradient Boosted Trees** demonstrated competitive performance, ranking as the second best-model and achieving strong results across all metrics. Interestingly, **Random Forest** underperformed compared to the single **Decision Tree**, despite being an ensemble method designed to improve upon individual tree predictions. This unexpected behaviour is likely due to several factors specific to its configuration. First, the averaging mechanism across multiple trees may lead to smoother probability estimates, potentially reducing prediction confidence and affecting calibration when individual trees are already well-tuned. Second, the random feature subset selection at each split may limit the model's ability to consistently exploit the most discriminative features. In contrast, boosting-based approaches, such as **Gradient Boosted Trees** and **XGBoost**, iteratively correct errors from previous trees, capturing complex decision boundaries and reducing bias, which explains their superior performance in this task.

To further investigate the models' discrimination capability, Receiver Operating Characteristic (ROC) curves for all models are shown in Figure 3.

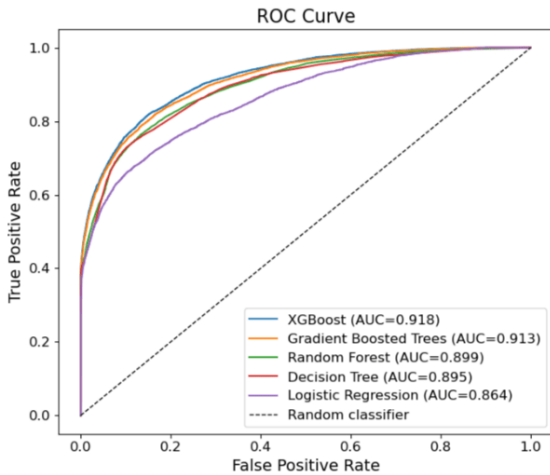


Figure 3: ROC Curves for all Models

The ROC analysis confirms the superiority of both **XGBoost** and **Gradient Boosted Trees**, with both

curves generally positioned above those of the other models. Moreover, as presented in Figure 4, to better distinguish between the top-performing models, a zoomed view of the ROC curve was generated, focusing on the operationally relevant region characterized by low false positive rates.

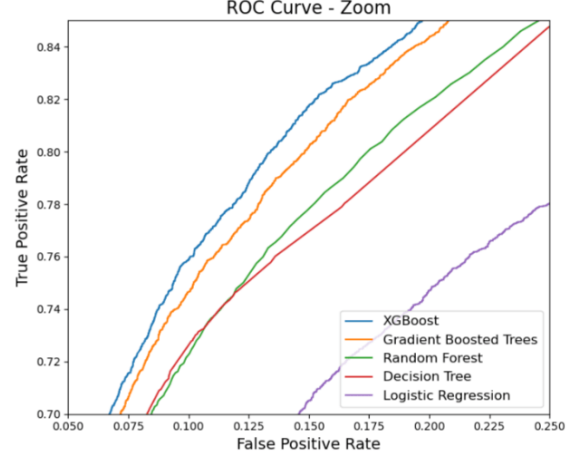


Figure 4: Zoomed ROC Curves for all Models

This view reveals subtle differences in model performance at application-specific thresholds. Indeed, **XGBoost** maintains a slight advantage over **Gradient Boosted Trees** in the low false positive rate region, which is particularly relevant for practical deployment scenarios where minimizing false positives is operationally important. In this context, the aim is to prevent flagging confirmed bookings as potential cancellations.

In summary, based on a comprehensive evaluation across multiple metrics and ROC analysis, **XGBoost** can be considered the optimal model for deployment, providing the most balanced performance in terms of overall accuracy, probability calibration, and discriminative ability.

6 Conclusion

This study addressed the problem of hotel booking cancellations by leveraging machine learning classification techniques applied to the Hotel Booking Demand dataset. The main objective was to develop a reliable predictive model capable of identifying reservations at high risk of cancellation, thus supporting more effective revenue management and operational decision-making in the hospitality sector.

A structured methodological workflow was adopted, starting from data exploration and preprocessing. Feature engineering transformations created more informative variables that better captured booking characteristics, while a systematic feature selection process identified the most relevant predictors and reduced model complexity. Then, five classification

algorithms with increasing levels of complexity were evaluated under a unified experimental framework. In particular, non-linear and ensemble-based models consistently outperformed the linear baseline, highlighting the presence of complex relationships between booking attributes and cancellations. In particular, **XGBoost** emerged as the best performing classifier, achieving the highest accuracy of 0.848, an F1-Score of 0.778, and the lowest log loss of 0.334, while also demonstrating superior discriminative ability, as confirmed by ROC analysis. Just behind **XGBoost**, **Gradient Boosted Trees** showed similar performance levels across all metrics.

Overall, the findings show that advanced boosting-based models are highly effective for predicting booking cancellations, offering both strong accuracy and well-calibrated probabilities. While the proposed approach proved effective, the analysis was limited to historical booking data and structured features derived from the available dataset. Future work could extend this framework by incorporating further temporal modelling strategies or additional contextual information, such as external or customer-level data, to enhance predictive performance and practical applicability [6].

References

- [1] D-EDGE. Hotel distribution report 2024: Have direct bookings reached their peak? Technical report, D-EDGE, 2024. Accessed: Dec. 22, 2025.
- [2] Nuno António, Ana de Almeida, and Luís Nunes. Hotel booking demand datasets. *Data in Brief*, 22:41–49, 2019.
- [3] Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. *An Introduction to Statistical Learning*. Springer, New York, 2013.
- [4] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001.
- [5] James Bergstra and Yoshua Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13:281–305, 2012.
- [6] Nuno António, Ana de Almeida, and Luís Nunes. Predictive models for hotel booking cancellation: a semi-automated analysis of the literature. *Tourism & Management Studies*, 15:7–21, 2019.